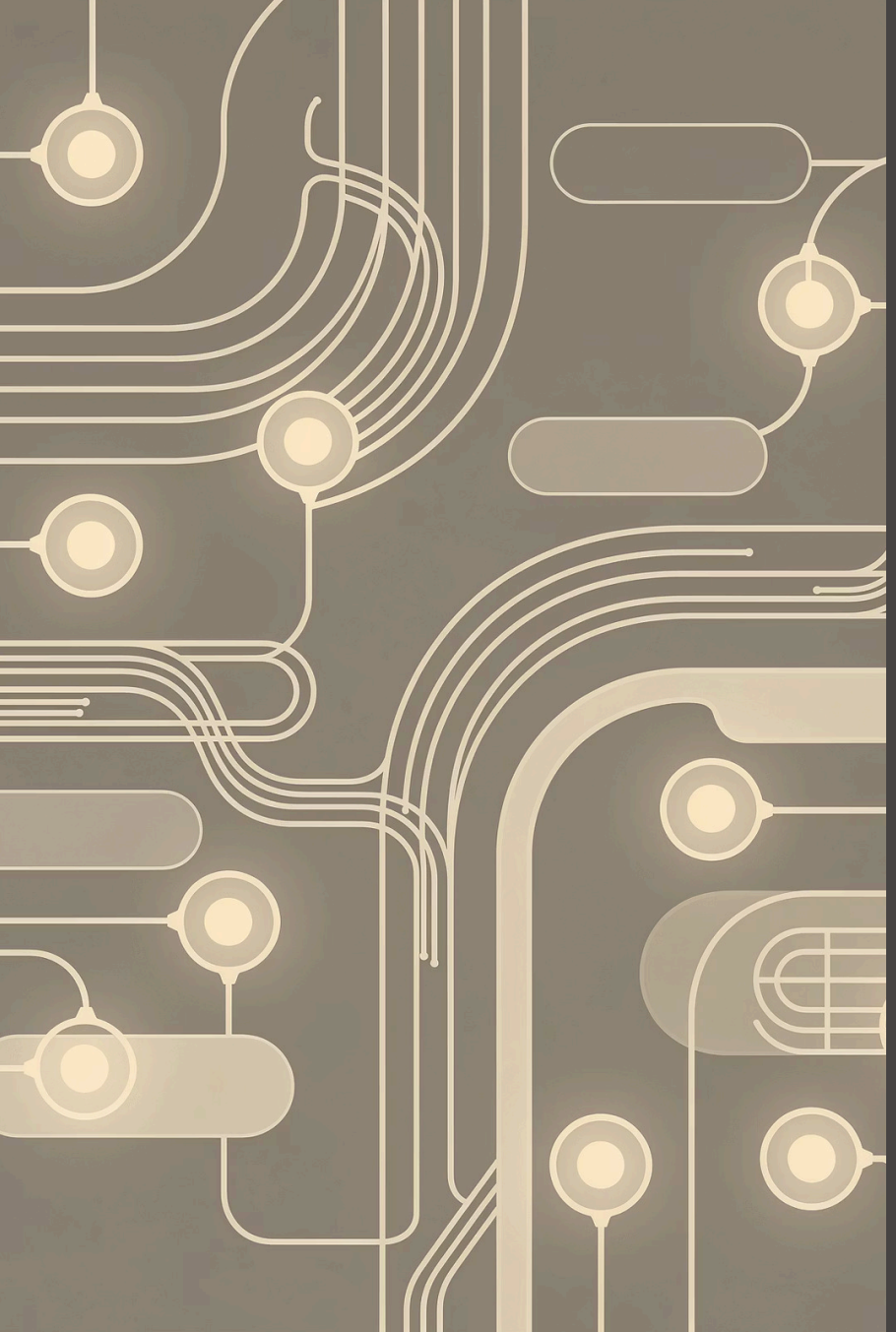


OpenShift SDN vs. OVN-Kubernetes: Evolving Cluster Networking

A deep dive into the networking models powering modern OpenShift clusters — from legacy SDN to the powerful, scalable OVN-Kubernetes architecture that defines the future.

OPENSHIFT NETWORKING

CNI DEEP DIVE

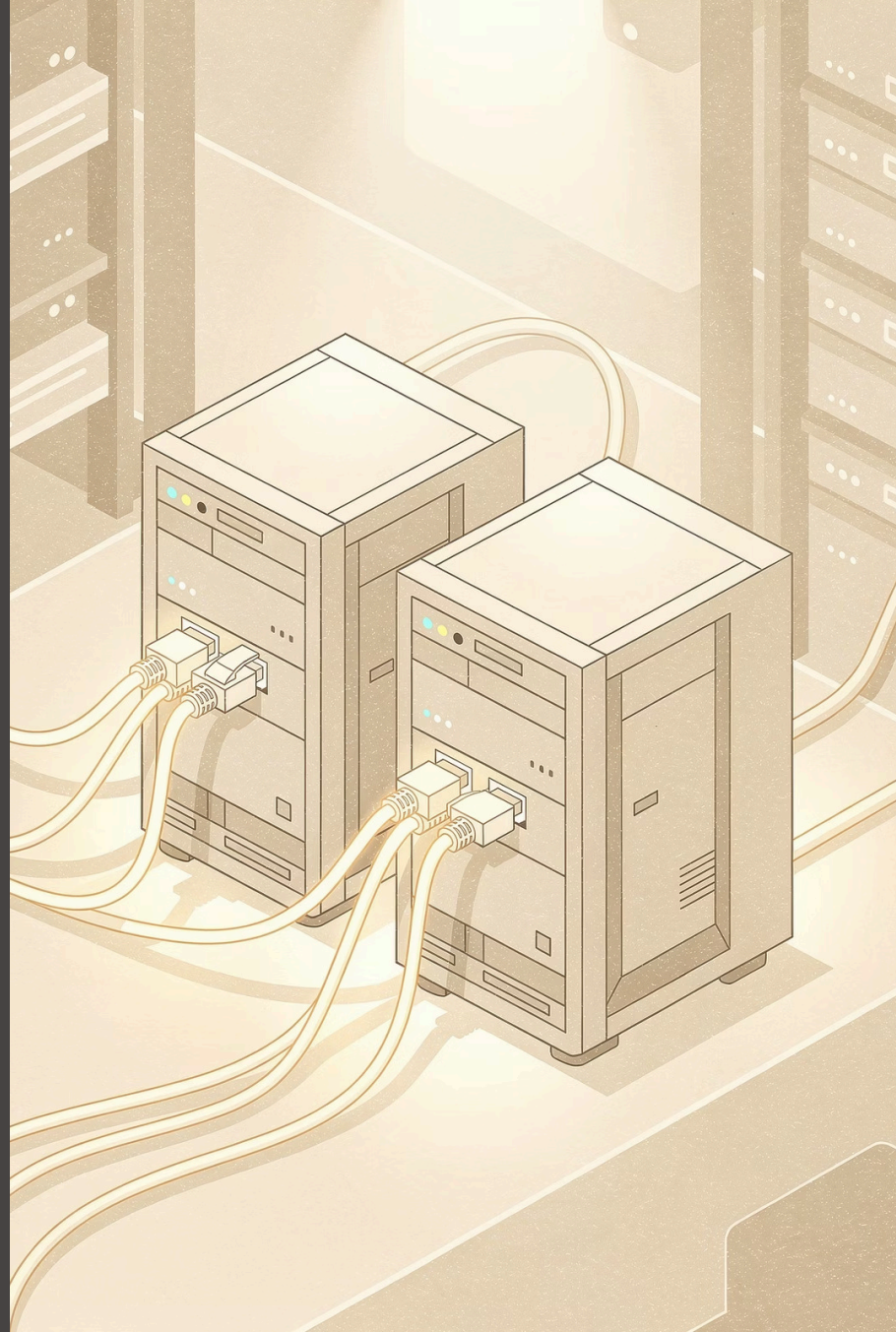
The Foundation: Pod-to-Pod Communication

OpenShift SDN

Historically provided basic pod-to-pod connectivity using **VXLAN overlay networks**. While functional for foundational workloads, it faced meaningful limitations in advanced feature support, network policy flexibility, and scalability at enterprise scale.

OVN-Kubernetes

The modern default CNI for OpenShift, built on **Open Virtual Network (OVN)** and **Open vSwitch (OVS)**. Utilizes **Geneve** for overlay networks, delivering enhanced performance, richer traffic control, and more sophisticated communication pathways between pods across nodes.



Service Networking: Connecting Your Applications

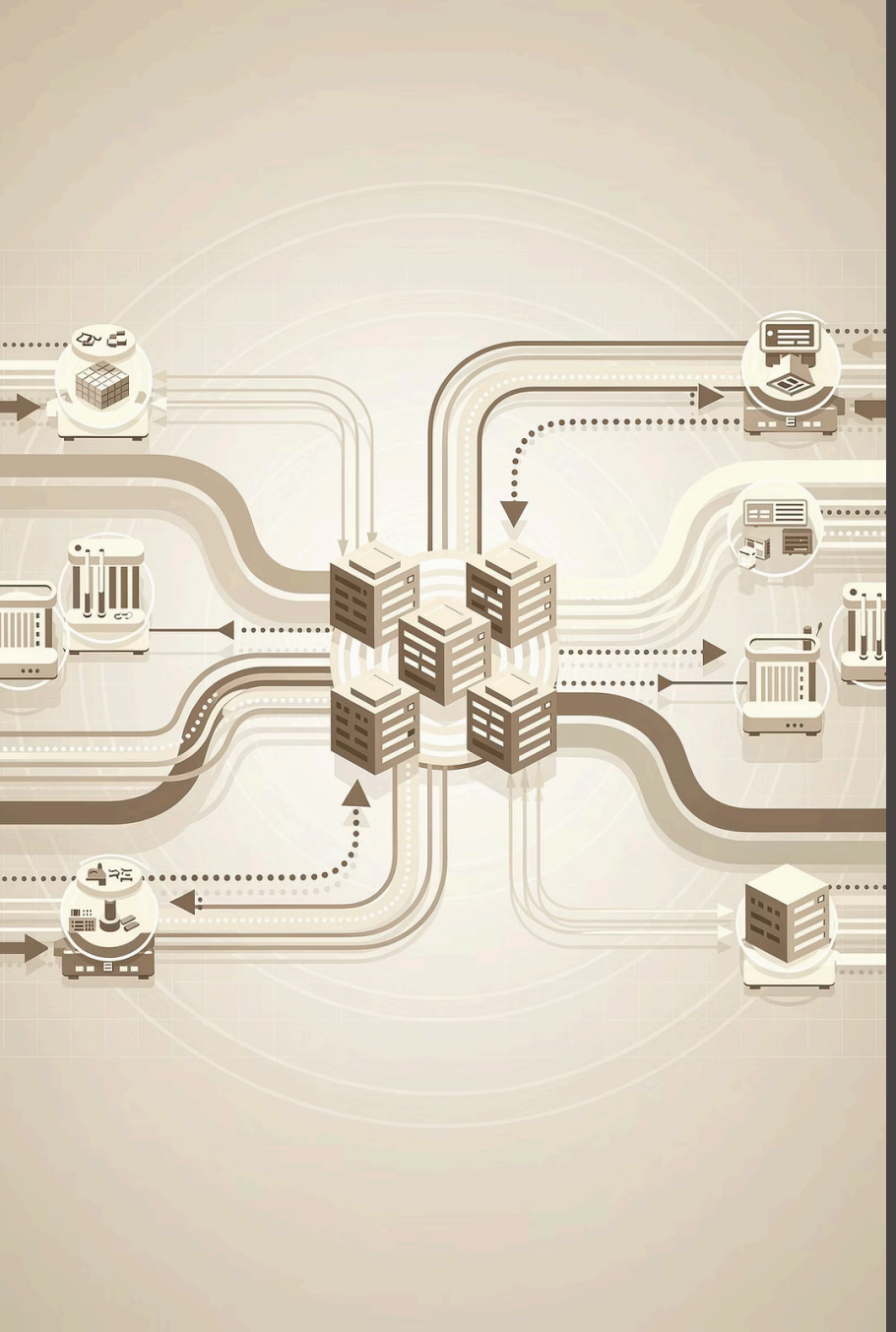
How each CNI handles Kubernetes Services — the backbone of internal load balancing and service discovery across your cluster.

OpenShift SDN

Supported standard Kubernetes Services for internal load balancing and service discovery. Traffic routing relied on **iptables rules** managed at the node level, which could become a bottleneck at scale with thousands of services and endpoints.

OVN-Kubernetes

Provides robust service networking through **OVN's distributed virtual routing and logical switches**. Supports advanced features including distributed DHCP and DNS resolution, ensuring seamless, low-latency service access. Traffic flows are managed programmatically at the OVN layer, dramatically improving performance and flexibility.



Ingress Controller Architecture: Traffic Entry Points

OpenShift SDN Approach

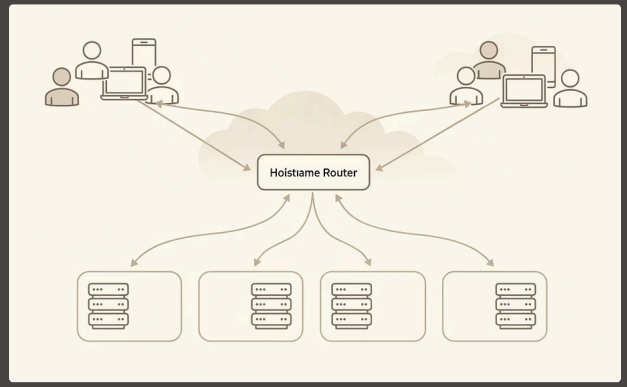
Typically relied on the **built-in OpenShift HAProxy router** acting as an ingress controller. This managed external access to services via a single, centrally deployed component, which — while effective — limited deployment flexibility and advanced traffic shaping options.

OVN-Kubernetes Approach

Integrates cleanly with standard **Kubernetes Ingress resources** and supports external Ingress controllers such as NGINX or Contour. OVN-Kubernetes manages the underlying network flows directly, providing a **flexible, programmable, and powerful** ingress path that scales with cluster complexity.

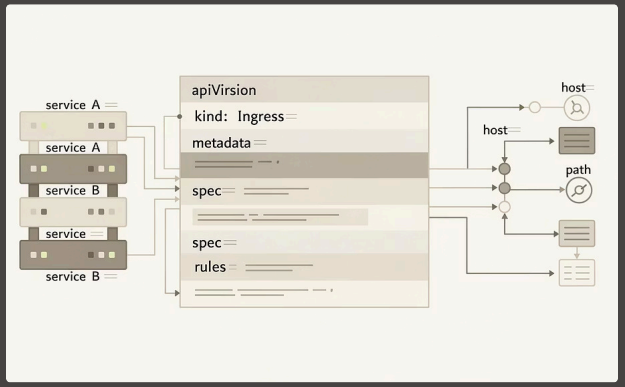
Routes vs. Ingress: OpenShift's Layer 7 Routing

OpenShift extends standard Kubernetes Ingress with its own **Route** resource — a higher-level abstraction purpose-built for enterprise traffic management.



OpenShift Routes

A native OpenShift abstraction providing **hostname-based routing**, TLS termination options, and fine-grained traffic controls. Routes are managed by the OpenShift router and offer a richer feature set than standard Ingress objects out of the box.



Kubernetes Ingress

The standard Kubernetes mechanism for HTTP/S routing into the cluster. OVN-Kubernetes supports both native Ingress resources **and** OpenShift Routes, ensuring that Layer 7 routing rules are efficiently enforced at the network layer regardless of which abstraction you choose.

TLS Termination: Securing Your Connections

OpenShift SDN

Supported TLS termination types **directly within the HAProxy router component**. Configuration was tightly coupled to the router, offering straightforward but less flexible certificate management and traffic handling.

OVN-Kubernetes

Facilitates TLS termination at the **Ingress controller or Route level**, decoupled from the underlying network fabric. OVN-Kubernetes ensures that all traffic — whether terminating at the edge or passing through to the pod — is handled securely and efficiently across distributed nodes.

TLS Termination Types Explained

OpenShift supports three distinct TLS termination strategies, each suited to different security and performance requirements. Choose based on your trust boundary and backend capabilities.

Edge Termination

TLS is **terminated at the ingress router**. The connection between the router and the backend pod is **unencrypted**. Best for trusted internal networks where simplicity and performance are priorities. Certificate management happens solely at the router.

Re-encrypt Termination

TLS terminates at the router, then a **new encrypted TLS connection** is established between the router and the destination pod. Provides **end-to-end encryption** while allowing the router to inspect and manage traffic. Requires a certificate on the backend pod.

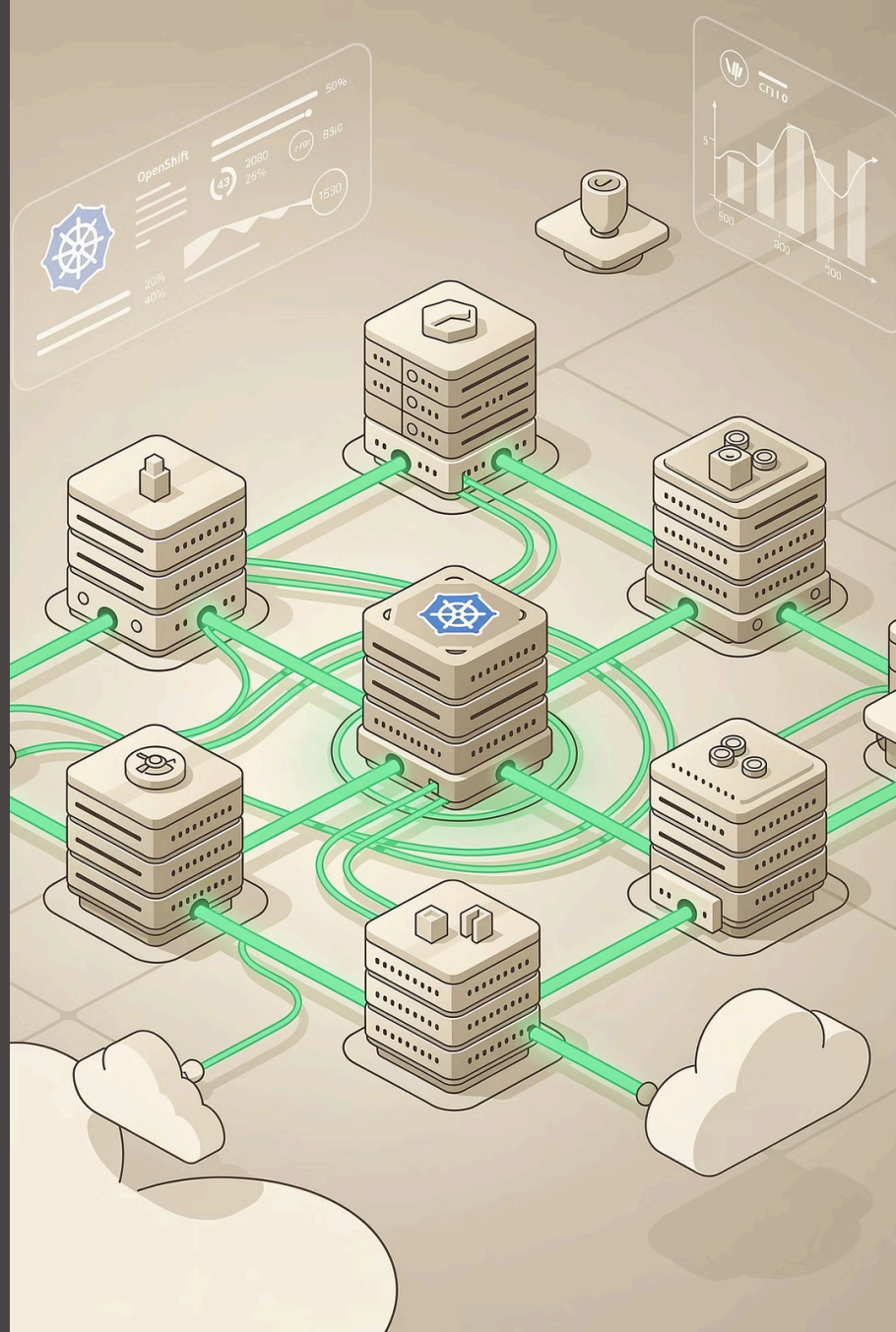
Passthrough Termination

TLS traffic is **not terminated at the router**. Encrypted traffic passes directly to the pod, which performs TLS termination itself. Ideal for applications requiring **mutual TLS (mTLS)** or when the application must own its own certificates completely.

OVN-Kubernetes: The Default and Future

> ⚠️ **Important Deprecation Notice:** OpenShift SDN is officially deprecated as of **OpenShift Container Platform 4.14** and is planned for removal in a future release. All new deployments should use OVN-Kubernetes.

OVN-Kubernetes is the **default CNI network provider** for all modern OpenShift deployments. Built on community-driven Open Virtual Network and Open vSwitch technologies, it is designed to handle the demands of large-scale, enterprise-grade containerized environments. Organizations still running OpenShift SDN should plan migration to OVN-Kubernetes to ensure continued support, access to new features, and long-term operational stability.



Key Advantages of OVN-Kubernetes

Scalability

Architected for **large-scale clusters** with thousands of nodes and services. Distributed control plane components eliminate single points of bottleneck, enabling predictable performance as workloads grow.

Advanced Network Policies

Supports **richer and more granular network policies**, including Admin Network Policies, distributed virtual routing, Egress IP, and Egress Firewall — capabilities unavailable or limited in OpenShift SDN.

Modern Protocol (Geneve)

Uses **Geneve** for overlay networking instead of VXLAN. Geneve supports extensible metadata in tunnel headers, offering greater flexibility for future protocol enhancements and multi-tenant environments.

Vendor-Agnostic Foundation

Built entirely on **community-developed OVN and Open vSwitch**, ensuring no vendor lock-in. Active upstream development means continuous improvements, broad ecosystem support, and alignment with industry standards.

Embracing the Future of OpenShift Networking

OVN-Kubernetes represents a **significant architectural evolution** in how OpenShift manages cluster networking. By moving from the legacy SDN model to a fully programmable, distributed virtual network fabric, OpenShift delivers the foundation your containerized applications need to thrive at scale.

🔒 Migrate Early

Plan your transition from OpenShift SDN to OVN-Kubernetes ahead of the removal deadline to avoid disruption.

⚡ Unlock Performance

Take advantage of distributed routing, advanced policies, and Geneve tunneling for superior throughput and reliability.

🌐 Future-Proof Your Stack

OVN-Kubernetes aligns with the upstream Kubernetes ecosystem, ensuring long-term support and continuous innovation.